

# Final Report

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Part IIA Project: *Neural Control with Adaptive State Estimation* (GG4)

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## Introduction

We are given a `Brain` and a Brain–Machine Interface (BMI), which controls an arm. We are tasked with controlling the position of the hand at the end of the arm. The `Brain` takes  $n_u = 2$  inputs and produces  $n_y = 16$  outputs; we assume it is a Linear Gaussian State–Space Model (LGSSM). The BMI takes the 16-dimensional observations from the `Brain` and applies a fixed linear decoder to map them to 2 latent states:  $x_1$ , the muscle selection rhythm; and  $x_2$ , power rhythm. The BMI stores a rolling history of  $x_1$ , which is passed through 4 fixed band–pass filters which measure how strongly  $x_1$  oscillates at that muscle’s defined excitation frequency. A history of  $x_2$  is also stored in state, and the current value is compared to a half– and full–period in the past: if strong oscillation is found, evidence accumulates; otherwise, evidence decays, producing a smooth power state. Muscle activation is the product of activation and the selection vector. Each muscle produces force proportional to its activation, and for each joint, the net force is the sum of signed muscle forces. Positive forces represent flexion; negative forces represent extension. A dry–friction threshold must be overcome, and once this is overcome, the remaining net force (once subtracting the dry friction) is divided by wet friction representing viscous resistance. Joint angles are integrated, and 2-link arm kinematics give the hand position.

## Rules of engagement

The holistic view of this project is to be able to reason and interpret a controller which takes observations from a brain (e.g. via electrodes on the scalp) to move a prosthetic arm. Critically, the brain is a black box and its model “parameters” cannot be directly observed. In the spirit of this view, we propose specific rules of engagement surrounding our analysis. We do not attempt to decompile the provided `Brain` executables to derive the system matrices. We do not average over multiple `Brain` seeds, removing noise to derive system matrices, we reserve this for evaluation purposes only to demonstrate robustness against noise. Where we do run experiments that are not physically realisable, we do this to demonstrate the direction of our analysis, and we replicate results in empirically valid ways alongside. We do not directly use the weights of the BMI in our control efforts: we allow ourselves to read their values for experimental design, but any results we must prove empirically before use. We only allow the use of the 16-dimensional

observations from the `Brain` and the hand position given by the BMI for closed-loop control; we do not rely on the internal state of the BMI (e.g., arm angles or latent state variables) for feedback. We also do not expect or attempt to control the elbow joint past  $180^\circ$  to mimic the capabilities of an anatomically accurate arm. We motivate these constraints as to achieve resilience and independence to both: inter-individual variation in brains (and therefore system matrices); and manufacturers of prosthetic arms (and therefore BMI decoding, state, and arm kinematics defined by model weights).

## System identification

As we assume the `Brain` is a LGSSM, we explore system identification as a means to understand the system matrices defining the model:

$$\begin{aligned} x_{t+1} &= \underline{A} x_t + \underline{B} u_t + w_t, & w_t &\sim \mathcal{N}(\underline{0}, \underline{Q}) \\ y_t &= \underline{C} x_t + o_t, & o_t &\sim \mathcal{N}(\underline{0}, \underline{R}) \end{aligned}$$

## Identification evaluation

To evaluate our estimates, we have two strategies: the reconstruction of observations and the reconstruction of frequency response. To measure the reconstruction of observations, we use the coefficient of determination defined with the Frobenius norm:

$$R^2 = 1 - \frac{\|Y - \hat{Y}\|_F^2}{\|Y - \bar{Y}\|_F^2}$$

This form is chosen for its effectiveness at aggregating errors across all time points and all observation dimensions. We weight this up against its disadvantages, which include ignorance of observation covariance and temporal correlations. To measure the reconstruction of frequency response, we inspect a frequency map derived analytically from estimates of system matrices, and compare characteristics with empirical results from a single run of the `Brain`. We find the transfer function of the LGSSM, dependent on only the deterministic components (matrices  $\underline{A}$ ,  $\underline{B}$ , and  $\underline{C}$ ):

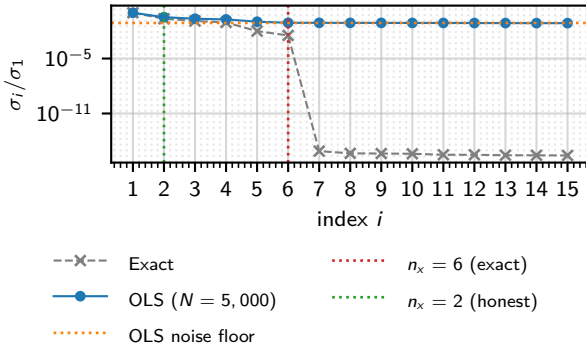
$$H(z) = \underline{C} (zI - \underline{A})^{-1} \underline{B}$$

We define an estimate to reconstruct the frequency response well if it can minimise the error between analytic and empirical values of  $|H(e^{j\omega})|$  across all neurons and frequencies  $[0, \pi]$ . Given the emphasis on

rhythm in the BMI we weigh frequency response reconstruction more than observation reconstruction if a compromise is necessary.

### Identification methods

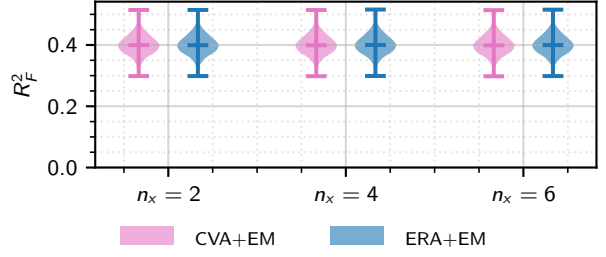
We run two Brain instances with and without input to isolate and subtract the noise. We can then calculate the exact Markov parameters, construct Hankel matrices, and form a singular-value spectrum. With this analysis we find  $\sigma_i/\sigma_1$  fall after  $i = 6$ , indicating the true latent dimension number to be  $n_x = 6$  (Figure 1). However, this method is not physically realisable: you cannot run the same physiological brain twice with the same random state. This would mean rewinding a brain, re-run a trial without a stimulus, and subtract: noise is truly stochastic outside of a simulation. A physically realisable single-run Ordinary Least Squares (OLS) regression estimate uncovers a noise floor that obscures modes 3–6. Balancing likelihood and model complexity with Bayesian Information Criterion (BIC) model selection, we identify  $n_x = 2 \neq 6$  (Figure 1).



**Figure 1:** Normalised Hankel singular values  $\sigma_i/\sigma_1$  for the  $40 \times 60$  block Hankel matrix ( $n_u = 2$ ,  $n_y = 16$ ). The exact spectrum is obtained by same-seed impulse subtraction, which is not physically realisable, and serves as a theoretical reference for the true system order. Modes 3–6 fall below the OLS noise floor of a 5000-step physically realisable recording; BIC model selection on the OLS estimates recovers  $n_x = 2$ . [Source](#)

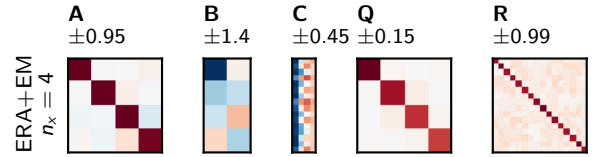
We plot a heatmap of  $|H(e^{j\omega})|$  and find that Canonical Variate Analysis (CVA) and Expectation-Maximisation (EM) to reconstruct  $u_1$  well, but not  $u_2$  over all  $n_x$  candidates  $\{2, 4, 6\}$  (Figure 4, N.B. only  $n_x = 4$  plotted). We turn to an estimation method that uses the Eigensystem Realisation Algorithm (ERA) and EM that utilises OLS-estimated Markov parameters and find similar results for  $n_x = 2$ : good reconstruction of  $u_1$  but poor for  $u_2$ . However, in contrast to CVA+EM, reconstruction of the  $u_2$  frequency response improved with  $n_x = 4$  and  $n_x = 6$  (Figure 4). More generally, we observe that the Brain acts as a low-pass filter. The observation reconstruction quality remained similar between CVA+EM and

ERA+EM (Figure 2).



**Figure 2:** Frobenius  $R_F^2$  of one-step-ahead Kalman predictions for CVA+EM and ERA+EM at  $n_x \in \{2, 4, 6\}$ , across  $N = 1000$  held-out trials of  $T = 2000$  steps (first 100 steps excluded as Kalman transient). Median  $R_F^2$ :  $n_x = 2$ : CVA+EM = 0.399, ERA+EM = 0.399;  $n_x = 4$ : CVA+EM = 0.399, ERA+EM = 0.400;  $n_x = 6$ : CVA+EM = 0.399, ERA+EM = 0.400. [Source](#)

As there was not a huge difference between  $n_x = 4$  and  $n_x = 6$ , we select ERA+EM and  $n_x = 4$ , which yields system matrices visualised in Figure 3.



**Figure 3:** Identified system matrices for ERA+EM ( $n_x = 4$ ). Red indicates positive, blue negative; colour scales are symmetric about zero. [Source](#)

### Closed-loop spectral control of brain observations

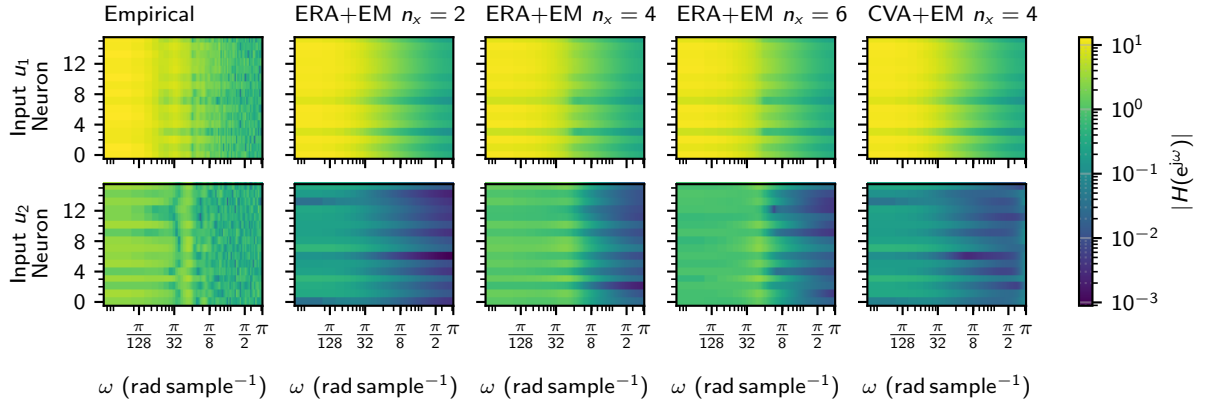
We test whether closed-loop stimulation can reproduce a prescribed frequency spectrum in neural population mean activity. We motivate the following evaluation framework and method with the emphasis on controlling rhythms in the Brain observations for influence over BMI latent variables and the arm downstream.

#### Spectral control evaluation

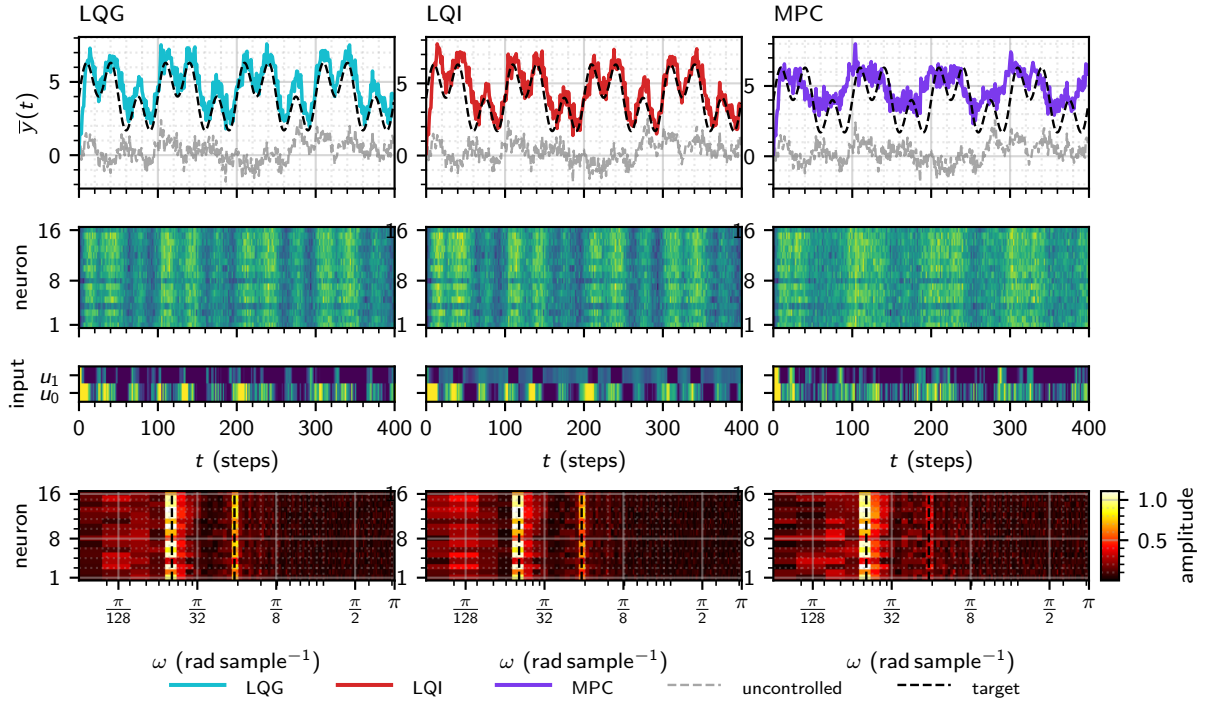
We use the following metrics to evaluate each controller: spectral Root Mean Square (RMS) error, which measures amplitude deviation at target frequencies; the maximum amplitude error, the worst-case per-component deviation; and the off-target power ratio, the mean spectral amplitude at non-target frequencies, relative to mean target amplitude.

#### Spectral control method

We offer 3 candidates for control strategies: Linear Quadratic Gaussian (LQG) to respect the Brain's LGSSM structure; Linear Quadratic Integral (LQI) to



**Figure 4:** Per-neuron frequency selectivity for inputs  $u_1$  (top row) and  $u_2$  (bottom row). Each panel shows  $|H(e^{j\omega})|$  for all  $n_y = 16$  neurons (vertical axis) across  $\omega \in [0, \pi]$  rad sample $^{-1}$  (horizontal axis), on a shared logarithmic colour scale. Leftmost column: empirical response (honest OLS estimate from a single 5000-step recording, DFT zero-padded to 512 points). Columns 2–4: analytic ERA+EM transfer function  $H(e^{j\omega}) = C(e^{j\omega}I - A)^{-1}B$  for  $n_x \in \{2, 4, 6\}$  respectively, showing the latent dimension sweep. Rightmost column: CVA+EM at  $n_x = 4$ , for estimator comparison at the chosen order. [Source](#)

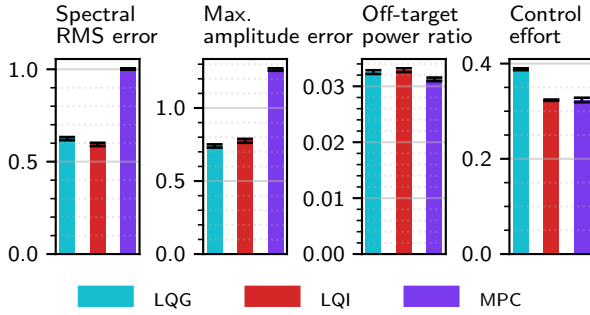


**Figure 5:** Closed-loop spectral control using the ERA+EM  $n_x = 4$  model ( $n_x = 4$ ), controller hyperparameters tuned by random search (8 trials each). The target is a sum of two sinusoids at  $f \in \{0.01, 0.03\}$  cycles per sample with amplitudes  $A \in \{1.50, 1.50\}$ , centred on a physical mean of 4. Each column shows one controller (LQG, LQI, MPC) with a per-neuron tracking cost ( $\lambda_{\text{var}} = 1$ ). Row 1: physical population mean  $\bar{y}(t)$ , uncontrolled baseline (grey dashed), and target reference (black dashed). Row 2: per-neuron observations  $y(t)$  as a neuron  $\times$  time heatmap. Row 3: applied inputs  $u(t)$ . Row 4: per-neuron amplitude spectral density (neuron  $\times$  angular frequency  $\omega$ , log scale); target frequencies shown as red dashed lines. [Source](#)

correct steady-state error; and Model Predictive Control (MPC) to optimise for and appreciate constraints on  $u_i \in [0, 1]$ .

### Spectral control results

Inspecting the raw population mean traces, we find LQG overestimates the population mean with a steady-state error, which is removed effectively by LQI. MPC fails to reconstruct the higher frequency components. We select LQI due to its lower control effort compared to LQG and better reconstruction performance over MPC.



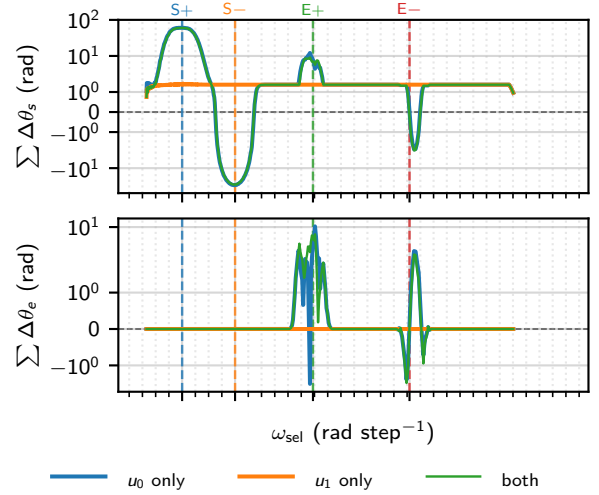
**Figure 6:** Spectral tracking metrics for LQG, LQI, and MPC controllers using the ERA+EM  $n_x = 4$  model ( $n_x = 4$ ). Each controller was tuned by random search (8 trials, seed 1) and evaluated over  $N = 20$  independent brain seeds. The target was a sum of sinusoids at  $f \in \{0.01, 0.03\}$  cycles per sample with amplitudes  $A \in \{1.50, 1.50\}$ , centred on a physical mean of 4. Bars show the mean; error bars show the standard error of the mean. Left: spectral RMS error (RMS amplitude deviation at target frequencies). Centre: maximum amplitude error (worst-case per-component deviation). Right: off-target power ratio (mean spectral amplitude at non-target frequencies relative to mean target amplitude). [Source](#)

## 1 Hand control

The mechanism of the BMI of muscle selection and a shared power state means that simultaneous control of different muscles is unlikely, as there is no granular control to direct different muscles at different power at the same time. This constrains our control strategy to be a sequential one, directing one muscle at a time.

### Muscle selection

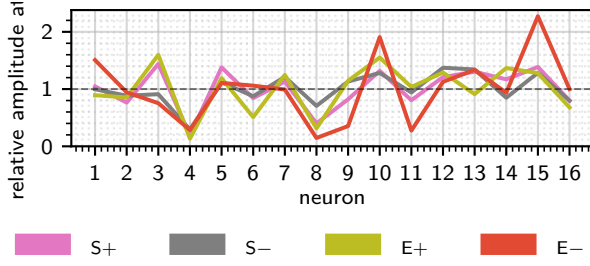
We sweep frequencies for both inputs and measure shoulder and elbow angles, determined by inverse kinematics to find the specific frequencies that excite specific muscles. We empirically recover the frequency bands defined in the BMI weights using a single seed, shown in TABLE and FIGURE.



**Figure 7:** Muscle-selection frequency sweep. At each of 512 logarithmically spaced angular frequencies  $\omega_{sel} \in [0.188, 2.639]$  rad step<sup>-1</sup>, open-loop sinusoidal drive (amplitude  $A = 0.45$ , centred at 0.5) is applied to  $u_0$  or  $u_1$  independently. The arm is observed via hand position and inverse kinematics. Top: signed cumulative shoulder displacement  $\sum \Delta \theta_s$ ; bottom: cumulative elbow displacement  $\sum \Delta \theta_e$ , each summed over the post-warmup window of 900 steps (symlog y-axis, linear threshold  $\pm 1$  rad); curves smoothed with a 5-sample moving average. Coloured vertical lines mark the four empirically identified band frequencies (S+ at  $\omega^* = 0.426$  rad step<sup>-1</sup>, S- at  $\omega^* = 0.780$  rad step<sup>-1</sup>, E+ at  $\omega^* = 1.301$  rad step<sup>-1</sup>, E- at  $\omega^* = 1.946$  rad step<sup>-1</sup>), identified from the open-loop  $u_0$  response by peak displacement magnitude (S $\pm$ ) and peak elbow selectivity  $|\Delta \theta_e| / (|\Delta \theta_e| + |\Delta \theta_s|)$  (E $\pm$ ). [Source](#)

We observe that amplitude decreases with frequency, making it harder to drive muscles that require higher frequencies. This may be due to the low-pass action of the Brain. We also observe cross-talk for elbow movements but not shoulder movements: the shoulder can be moved independently of the elbow, but the elbow cannot be moved independently of the shoulder. This motivates a control strategy that first achieves the correct elbow angle, then disregards control over the elbow, and uses S+ and S- only to correct the shoulder angle. In addition, during the initial correction angle phase, we should alternate between correcting the elbow angle and correcting the shoulder angle, as attempting to move the elbow moves the shoulder.

We then ask whether neural observations carry a muscle-specific spatial code: if different neurons respond preferentially at each selection frequency, the control target could be weighted to favour one muscle over another. For each selection frequency we record the full  $n_y = 16$  D amplitude spectrum and normalise the per-neuron profile by its own neuron-mean (Figure 8).



**Figure 8:** Neural correlates of muscle selection. For each of the four selection frequencies identified in Figure 7, a sustained open-loop sinusoidal trial ( $u_0 = 0.5 + 0.4 \sin(2\pi f^* t)$ ,  $u_1 = 0.5$ ) recorded  $n_y = 16$  D neural observations over 1024 steps following 500 burn-in steps. Each line shows the amplitude at  $\omega_{\text{sel}}^*$  per neuron, normalised by the muscle's own neuron-mean; the dashed line marks the mean ( $= 1$ ). Profiles that collapse onto each other confirm that every muscle recruits the same relative neural activation pattern, so the  $16 \text{ D} \rightarrow x_1$  projection is not muscle-specific. [Source](#)

Profiles are broadly consistent at the two shoulder frequencies and diverge at higher elbow frequencies, where the low-pass action of the Brain attenuates neurons differentially (Figure 4). Muscle identity is therefore encoded temporally by the oscillation frequency of  $x_1$ , not spatially by which neurons are active. No differential neural weighting offers selectivity beyond what the BMI's bandpass filters already provide, reinforcing the sequential, frequency-based control strategy described above.